

e-ERFA

Project Proposal

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1. Introduction

e-ERFA is a modern digital platform that redefines the way scholarships and financial assistance are managed. The platform introduces an online portal that enables students to submit scholarship applications with ease, while providing the ERFA Department with a structured and transparent system for managing, verifying, and analyzing submissions. It eliminates the administrative burden of manual processes by establishing a streamlined workflow, allowing students to apply smoothly and monitor their application status.

2. Objective

The platform streamlines the scholarship process by providing a user-friendly interface for submitting applications and receiving ERFA-related notifications. Simultaneously, it offers ERFA officers an efficient way to manage applications through tools for document review, data validation, and application analysis.

3. Problem Description

What

Current scholarship and financial assistance processes are often resource-intensive and reliant on manual administrative workloads. This approach leads to several systemic challenges, including a lack of real-time access to information about applicants. This results in administrators being unable to retrieve and validate student data efficiently, and forces students to make frequent in-person visits to clarify document requirements or confirm their application status. The reliance on physical documentation can also introduce vulnerabilities to data consistency and processing errors inherent in manual handling.

Why

A modernized online platform is essential to address these operational inefficiencies and elevate the administrative workloads. By providing a single point of access, the solution ensures a transparent and streamlined experience for all users. The portal will not only digitalize key workflows but also establish a foundation for enhanced data management and reporting. Ultimately, the digitization of the entire process is crucial to transition from a manual, reactive model to a proactive, data-driven framework that supports long-term administrative needs.

4. Target Industry

- Education
- Government & Public Sector
- Finance
- Non-Profit Organizations and Foundations

5. Methodology

The e-ERFA platform will be developed using a hybrid Agile approach (composition of Iterative and Agile methodology), leveraging **MERN stack** to ensure a rapid, iterative, and scalable development process. Our methodology is designed to tackle the identified problems systematically, focusing on a component-based architecture for both the frontend and backend.

- **Frontend Development:** The user interface will be built using **React.js**. This component-based approach will enable us to create a highly responsive and modular design, making it easier to manage and update both the student and ERFA officer portals.
- **Backend Development:** The server-side logic will be powered by **Node.js with Express.js**. This JavaScript-based environment ensures a unified stack and facilitates the development of robust APIs that handle all application requests, including authentication, data retrieval, and automated processes.
- **Data Management:**
 - **MongoDB** will serve as the primary database for storing application data, user profiles, and announcements. Its flexible schema will be particularly beneficial for accommodating evolving form requirements.
 - **LangChain** will be integrated to orchestrate the backend for the ERFA Assist Chatbot. It will connect the chatbot to a **Vector Database** (Pinecone) for efficient semantic search of FAQ and policy data, enabling the chatbot to provide accurate and context-aware responses.
- **Document Processing:** The document upload module will use **Tesseract.js** or **OpenCV** to perform OCR and image processing. This allows the system to validate documents, reducing the manual effort of the ERFA officers.
- **Automation:**
 - **Email Triggers:** We will implement an email API (Nodemailer or a third-party service) to send automated notifications to students for updates, interview schedules, and application status changes.

- **Interview Scheduling:** The system will manage interview time slots, ensuring a fair and conflict-free scheduling process for both students and faculty.

6. Project Scope

Included:

- Development of a dual-portal system for students and ERFA officers.
- Implementation of the student portal, including authentication restricted to official domain emails, an application dashboard, and a home page for announcements.
- Integration of an OCR-based document upload module for digital form and document submission.
- Development of the ERFA Officer portal with an analytics dashboard for key statistics, scholarship application filters, and profile search functionality.
- Creation of the ERFA Assist Chatbot to provide real-time guidance and answer FAQs.
- Functionality for officers to approve, reject, or return applications and post scholarship updates via email triggers.
- Implementation of a system-based interview scheduler and a corresponding ERFA interview panel for interviewers.

Limitations:

- Development of the data required for the chatbot, such as FAQs and scholarship policies, is outside the scope of the project and is assumed to be provided by the ERFA Department.
- The system is developed for a single institution and does not include multi-institutional support or APIs for external third-party integrations.
- The final decision to approve or reject an application remains with the ERFA officers; the system only facilitates the review and provides data for decision-making.

Assumptions:

- The system's successful operation is contingent upon users having a stable internet connection.
- The project assumes that the ERFA Department will provide the necessary data for the chatbot's functionality.
- The project assumes the cooperation of ERFA officers and faculty for the manual components of the scholarship process, such as physical document verification and conducting interviews.

7. Feasibility Study

The feasibility of developing the e-ERFA platform is contingent upon a strategic assessment of key factors, including technical complexity, resource allocation, and operational risks. The project is considered highly feasible due to its defined scope and the availability of established technologies to address the identified challenges.

i. Risks Involved

- **Technical Complexity:** The primary technical risk lies in the accurate and reliable implementation of the OCR-based document upload module and the ERFA Assist Chatbot. The OCR system must accurately parse diverse document formats, while the chatbot must provide precise, real-time responses based on structured data.
 - **Mitigation:** This risk will be mitigated by leveraging industry-standard libraries and APIs for OCR (Tesseract or Cloud Vision AI) and natural language processing (NLP). To ensure reliability and accuracy, the chatbot's responses are based on data provided by ERFA, avoiding the unpredictable nature of complex AI models.
- **User Adoption:** There is a potential risk that ERFA officers and faculty may face resistance to transitioning from a manual, paper-based system to a fully digital platform. This could hinder the system's intended efficiency gains.
 - **Mitigation:** The project will incorporate a few training sessions for the ERFA officers. The system's user interface is designed for simplicity and will be continuously refined based on user feedback to ensure a seamless and intuitive experience.

ii. Resource Requirements

- **Technical Resources:**
 - **Development Stack:** The project will be built using the MERN stack (MongoDB, Express, React, Node.js), which is a modern, scalable, and well-supported technology stack.
 - **OCR & Chatbot APIs:** Integration with third-party services or open-source libraries for OCR and chatbot functionality will be necessary.

- **Email Services:** A dedicated service, such as Mailchimp or a direct SMTP (Simple Mail Transfer Protocol) provider, will be integrated to handle automated email notifications for scholarship updates, interview scheduling, and application status changes.

8. Solution Application Areas

The application is designed for the Education Sector to digitize and streamline the scholarship application process, making it a centralized, paperless, and efficient platform for both administrators and students. The solution aims to ease the administrative workload, enhance operational efficiency, and provide a transparent experience for students.

9. Tools/Technology

- **Frontend:** React.js for building interactive user interfaces.
- **Backend:** Node.js with Express.js for server-side development.
- **Database:** MongoDB for efficient data storage and retrieval.
- **Chatbot:** LangChain for building the ERFA Assist Chatbot and Pinecone vector DB for semantic search and retrieval in the chatbot.
- **Chatbot:** Tesseract.js/OpenCV/Cloud Vision AI services for OCR-based document validation.
- **Version Control:** Git with platforms like GitHub for collaborative code management.

10. Expertise of the Team Members

Our team has the complementary skills needed for the e-ERFA project. We have a background in web development and database management from university courses and external bootcamps. One member has experience in chatbot development, and another in email services, both of which are crucial for this project. We have also collaborated on projects involving OCR technology.

11. Milestones

1. Project Planning (Weeks 1-2)

- Roles Involved: All

- Tasks:
 - Requirement Analysis
 - Project Scope Definition
 - Feasibility Study and Project Timeline

2. **Design Phase (Weeks 3-7)**

- Roles Involved: All
- Tasks:
 - System Architecture Design
 - UI/UX Design
 - Database Schema Design

3. **Development Phase (Weeks 9-30)** This phase is divided into two parts, corresponding to the features to be developed in FYP-I and FYP-II.

- User Management Features
- Digital Scholarship Form
- Scholarship Post Feed
- Advanced Search for students' profiles
- ERFA Assist Chatbot
- ERFA Email Updates
- OCR-Based Document Upload
- Applicant Assessment & Approval System
- Reporting and Analytics Dashboard
- Interview Scheduling
- ERFA Interview Panel
- Batch Upload Scholarship Results

Features: FYP-I

In FYP-I, the primary focus will be on developing the core user interfaces, database connectivity, and essential functionalities for both the student and ERFA officer portals to bring the system into operation.

- User Management Features
- Creation of Digital Scholarship Form
- Scholarship Post Feed
- Advanced Search for students' profiles

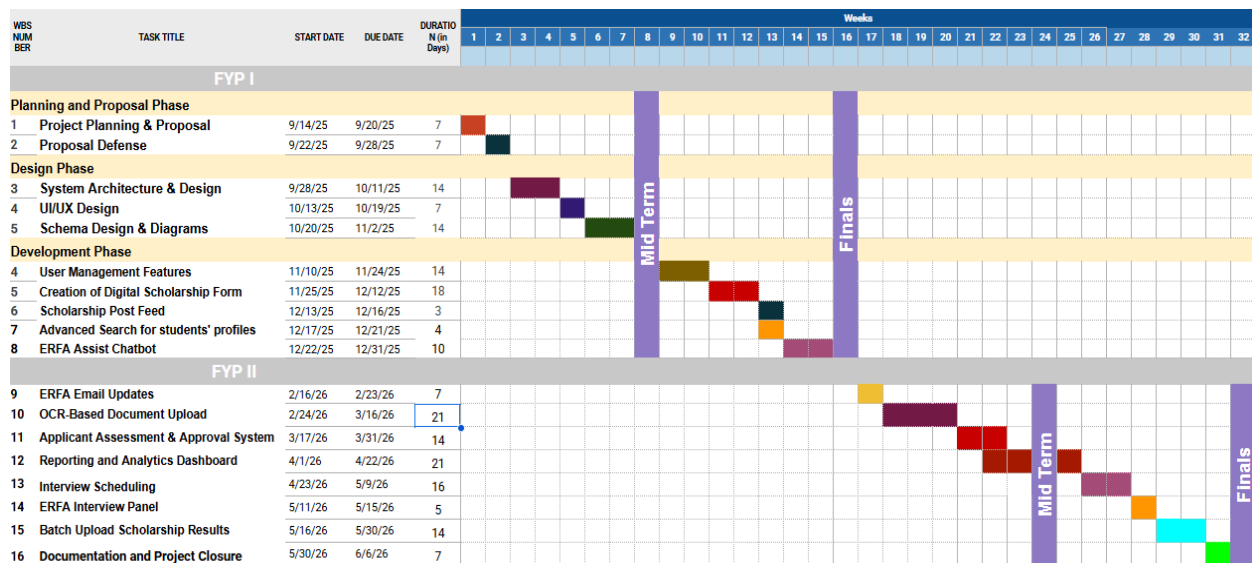
- ERFA Assist Chatbot

Features: FYP-II

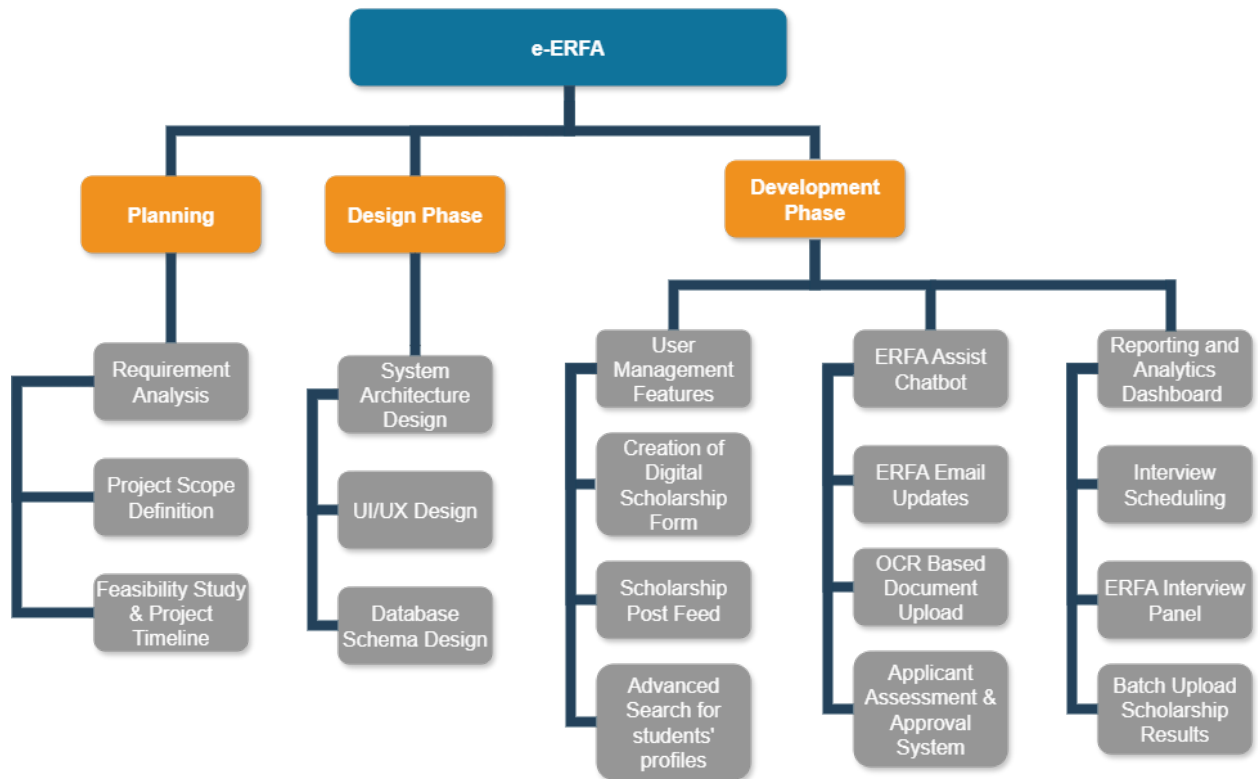
In FYP-II, the focus shifts to building advanced features to enhance the system's efficiency and transparency.

- ERFA Email Updates
- OCR-Based Document Upload
- Applicant Assessment & Approval System
- Reporting and Analytics Dashboard
- Interview Scheduling
- ERFA Interview Panel
- Batch Upload Scholarship Results

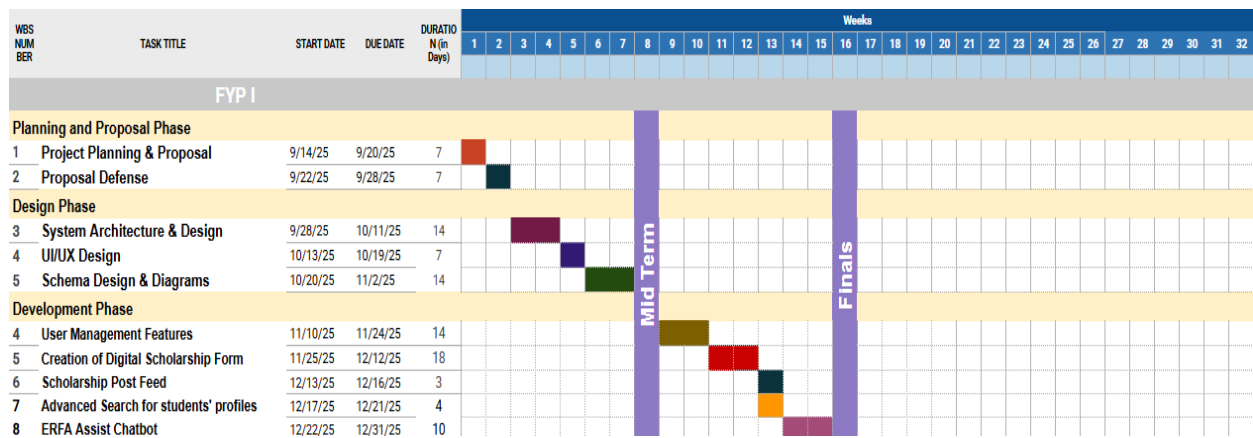
12. Project Schedule



13. Work Breakdown Structure (WBS)



FYP 1:



FYP 2:

WBS NUM BER	TASK TITLE	START DATE	DUE DATE	DURATIO N (in Days)	Weeks																															
					1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
FYP II																																				
9	ERFA Email Updates	2/16/26	2/23/26	7																																
10	OCR-Based Document Upload	2/24/26	3/16/26	21																																
11	Applicant Assessment & Approval System	3/17/26	3/31/26	14																																
12	Reporting and Analytics Dashboard	4/1/26	4/22/26	21																																
13	Interview Scheduling	4/23/26	5/9/26	16																																
14	ERFA Interview Panel	5/11/26	5/15/26	5																																
15	Batch Upload Scholarship Results	5/16/26	5/30/26	14																																
16	Documentation and Project Closure	5/30/26	6/6/26	7																																

14. References

[1] “Tesseract OCR Documentation.” Google, Last Accessed Sept 2025.

<https://github.com/tesseract-ocr/tesseract>

[2] Doe, Jane. “Automating Scholarship Management Systems.” *Journal of Educational Technology*, June 2021, pp. 35–49.